

**UNIVERSITY OF TRANSPORT AND COMMUNICATIONS
FACULTY OF INFORMATION TECHNOLOGY**

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**COURSE PROJECT REPORT
SOFTWARE ENGINEERING**

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Group 9: STUDENT MANAGEMENT SYSTEM

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TABLE OF CONTENTS

1. Functional Requirements.....	4
1.1. Login and Account Management.....	4
1.2. Student Information Management	4
1.3. Class Management	4
1.4. Subject Management	5
1.5. Grade Management	5
1.6. Academic Result Management.....	6
1.7. Statistics and Reports	6
1.8. Logout	6
2. Non-Functional Requirements	7
2.1. Usability	7
2.2. Reliability	7
2.3. Performance	7
2.4. Security.....	7
2.5. Development	8
3. Data Flow Diagram	9
3.1. Level 0.....	9
3.2. Level 1	10
4. Use case Diagram.....	11
4.1. List of Actors.....	12
4.2. List of Use Cases	12
4.3. Use Case Specifications	13
4.3.1. Login.....	13
4.3.2. Change Password.....	14
4.3.3. Logout.....	14
4.3.4. View Personal Information.....	15
4.3.5. View Grades	16
4.3.6. View Academic Results	16
4.3.7. View Assigned Classes	17
4.3.8. View Students by Class	17
4.3.9. View Assigned Subjects.....	18
4.3.10. Enter Grades.....	18

4.3.11. Update Grades	19
4.3.12. Manage Students.....	19
4.3.13. Manage Classes	21
4.3.14. Manage Subjects.....	21
4.3.15. View Statistics	22
4.3.16. Generate Reports.....	22
5. Class Diagram.....	23
6. Data Model.....	24
7. Interface Design Description	25
7.1. List of Screens	25
7.2. Screen Details.....	26
7.2.1. Login.....	26
7.2.2. Home	26
7.2.3. Student Management.....	28
7.2.4. Class & Subject Management.....	29
7.2.5. Grade Management	30
7.2.6. Academic Results	31
7.2.7. Statistics & Reports.....	32
7.2.8. Account Management.....	33

1. Functional Requirements

The Student Management System is developed to support the school in managing student information, classes, subjects, grades, and academic results.

1.1. Login and Account Management

- + The system allows Student, Teacher, and School Administration to log in to the system using an account and password.
- + The system requires users to provide a username and password.
- + The system validates the login information.
- + If the login information is correct, the system allows the user to access functions corresponding to their role.
- + If the login information is incorrect, the system displays an error message.
- + The system does not allow unauthenticated users to access functions that require authentication.
- + Users can change their account password.

1.2. Student Information Management

- + School Administration can manage student information in the system.
- + School Administration can view the list of students.
- + School Administration can add new student information.
- + School Administration can update student information.
- + School Administration can delete student information.
- + The system validates the information before adding or updating student information.
- + The system saves the student information after the operation is completed successfully.
- + Student can view their personal information.
- + Teacher can view information of students belonging to the classes they are responsible for.

1.3. Class Management

- + School Administration can manage class information in the system.
- + School Administration can view the list of classes.

- + School Administration can add a new class.
- + School Administration can update class information.
- + School Administration can delete a class.
- + The system stores information about students belonging to each class.
- + Teacher can view the list of classes they are assigned to.
- + Teacher can view the list of students belonging to the classes they are responsible for.

1.4. Subject Management

- + School Administration can manage subject information in the system.
- + School Administration can view the list of subjects.
- + School Administration can add a new subject.
- + School Administration can update subject information.
- + School Administration can delete a subject.
- + Teacher can view the list of subjects they are assigned to.
- + The system stores subject information and the relationships between subjects, classes, and students.

1.5. Grade Management

- + Teacher can manage the grades of students belonging to the classes and subjects they are assigned to.
- + Teacher can enter grades for students.
- + Teacher can view students' grades.
- + Teacher can update students' grades.
- + The system validates the grade value before saving it.
- + The system does not save the grade if the grade value is invalid.
- + The system stores the grade corresponding to each student and subject.
- + Student cannot modify their own grades.
- + School Administration can view grade information for management purposes.

1.6. Academic Result Management

- + Student can view their academic results.
- + Student can view their grades for each subject.
- + Student can view their overall academic results by semester.
- + Student can only view their own academic results.
- + Teacher can view the academic results of students belonging to the classes they are responsible for.
- + School Administration can view students' academic results for management purposes.

1.7. Statistics and Reports

- + School Administration can use the statistics and academic reports functions.
- + School Administration can view the number of students by class.
- + School Administration can view academic results by class.
- + School Administration can view academic results by subject.
- + School Administration can request the system to generate academic statistics by semester.
- + The system displays statistical results in an appropriate data format.
- + The system provides reports based on the data stored in the system.

1.8. Logout

- + Student, Teacher, and School Administration can log out of the system.
- + When the user logs out, the system terminates the current login session.
- + After logging out, the user must log in again to use functions that require authentication.

2. Non-Functional Requirements

2.1. Usability

- + The system must provide an interface appropriate for each role: Student, Teacher, and School Administration.
- + The system must display functions corresponding to the permissions of each role.
- + The system must display a notification when the user enters invalid data.
- + The system must display a confirmation message after successfully adding, updating, or deleting data.
- + Required information fields must be clearly marked on the interface.
- + The main functions must be organized into groups corresponding to each role.

2.2. Reliability

- + The system must maintain stable operation under normal operating conditions.
- + The system shall provide reliable operation during the specified operating period, with minimal system failures.
- + The system must not lose grade data when an error occurs during the update process.
- + If a data update operation fails, the system must retain the data from before the operation.
- + The system must display an error message when an error occurs during data processing.
- + The system must perform periodic data backups to reduce the risk of data loss.

2.3. Performance

- + The system must respond to user operations within an appropriate time under normal operating conditions.
- + The system must return login results within 3 seconds.
- + The system must display the list of students within 3 seconds.
- + The system must process grade entry or grade update operations within 3 seconds.

2.4. Security

- + The system must control access permissions based on the user's role.
- + Student is only allowed to view their own personal information and academic results.

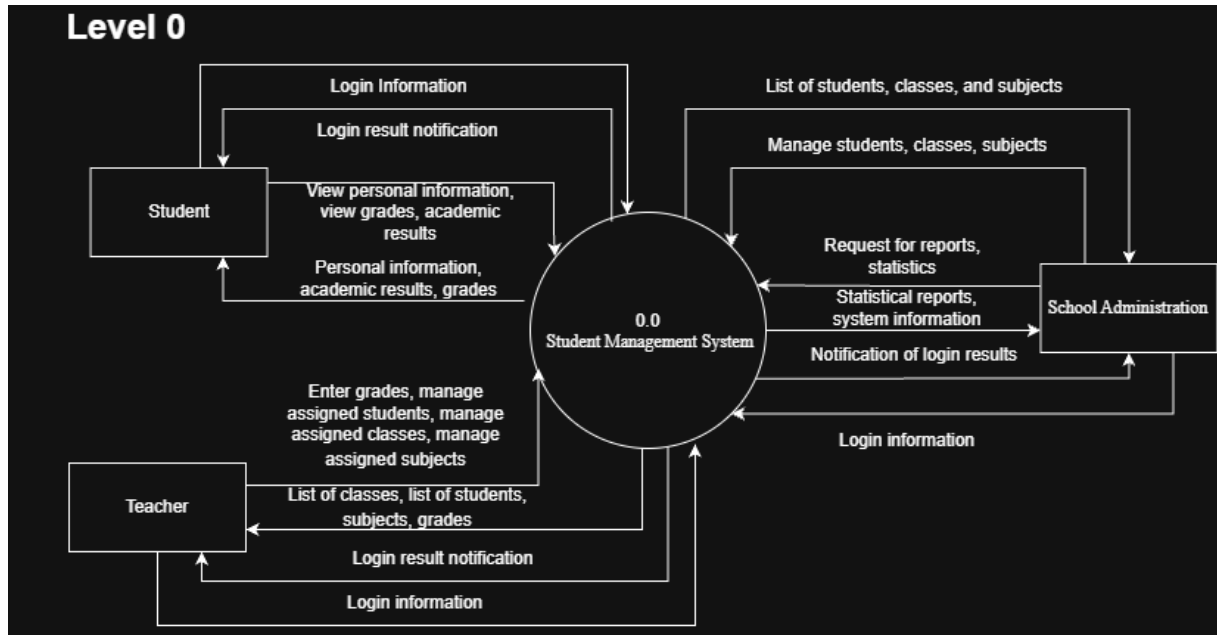
- + Teacher is only allowed to manage grades for the classes and subjects they are assigned to.
- + School Administration is allowed to perform management functions according to the assigned permissions.
- + Users must log in before accessing functions that require authentication.
- + The system must protect user passwords using an appropriate encryption or hashing method.
- + The system must prevent unauthorized access to data.
- + The system must have measures to prevent common security vulnerabilities such as SQL Injection and Cross-Site Scripting (XSS).

2.5. Development

- + The system must be developed using Kotlin and C++ as the programming languages.
- + Android Studio must be used as the primary development environment for the Kotlin-based application.
- + Visual Studio Code must be used as the development environment for the C++ application.
- + The system must use SQL for database management and data storage.
- + The system must be developed based on the agreed Functional Requirements.
- + The system must be developed based on the agreed Functional Requirements.

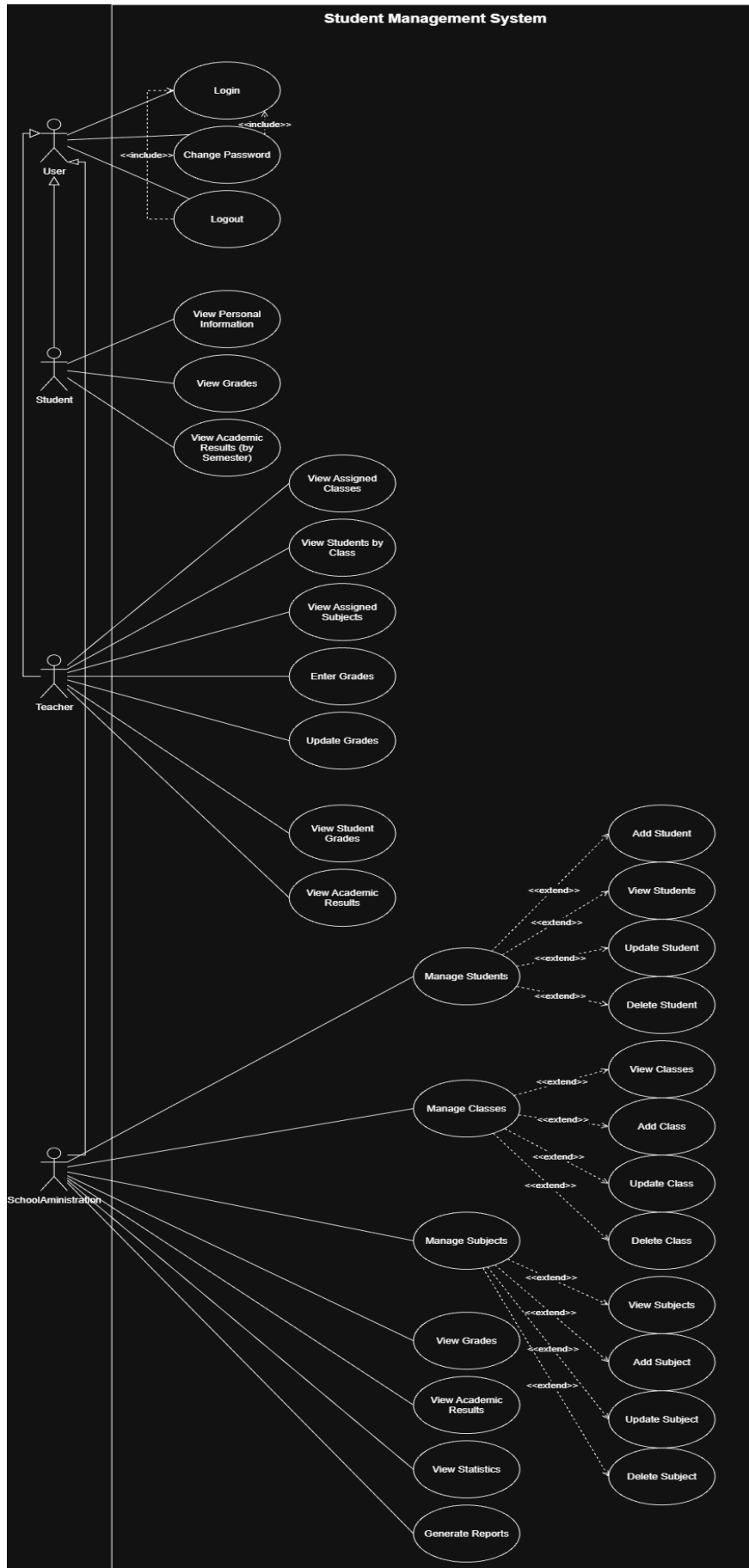
3. Data Flow Diagram

3.1. Level 0





4. Use case Diagram



4.1. List of Actors

	Actor	Description
1	User	General actor representing users who interact with the system
2	Student	Can log in, view personal information, grades, and academic results
3	Teacher	Can view assigned classes/subjects, view students, and manage students' grades
4	School Administration	Can manage students, classes, subjects, view grades/results, and generate statistics and reports

4.2. List of Use Cases

	Use Case
1	Login
2	Change Password
3	Logout
4	View Personal Information
5	View Grades
6	View Academic Results
7	View Assigned Classes
8	View Students by Class
9	View Assigned Subjects
10	Enter Grades
11	Update Grades
12	View Student Grades
13	View Academic Results
14	Manage Students
15	Add Student

	Use Case
16	View Students
17	Update Student
18	Delete Student
19	Manage Classes
20	View Classes
21	Add Class
22	Update Class
23	Delete Class
24	Manage Subjects
25	View Subjects
26	Add Subject
27	Update Subject
28	Delete Subject
29	View Grades
30	View Academic Results
31	View Statistics
32	Generate Reports

4.3. Use Case Specifications

4.3.1. Login

Attribute	Description
Use Case	Login
Actor	User
Goal	Allow Student, Teacher, and School Administration to access the system

Attribute	Description
Precondition	The user has a registered account
Trigger	The user enters a username and password and selects Login
Main Flow	1. User enters username and password 2. System validates the login information 3. System identifies the user's role 4. System grants access to functions according to the user's role
Alternative Flow	If the username or password is incorrect, the system displays an error message and asks the user to try again
Postcondition	The user is successfully logged in and a login session is created

4.3.2. Change Password

Attribute	Description
Use Case	Change Password
Actor	User
Goal	Allow the user to change their account password
Precondition	The user is logged in
Main Flow	1. User selects Change Password 2. User enters the current password 3. User enters a new password 4. System verifies the current password 5. System updates the password 6. System displays a success message
Alternative Flow	If the current password is incorrect, the system displays an error message. If the new password is invalid, the system asks the user to enter it again
Postcondition	The account password is successfully updated

4.3.3. Logout

Attribute	Description
Use Case	Logout

Attribute	Description
Actor	User
Goal	End the user's current session
Precondition	The user is logged in
Main Flow	1. User selects Logout 2. System confirms the logout request 3. System terminates the current session 4. System redirects the user to the Login screen
Alternative Flow	User cancels the logout operation → the current session remains active
Postcondition	The user is successfully logged out

4.3.4. View Personal Information

Attribute	Description
Use Case	View Personal Information
Actor	Student
Goal	Allow a student to view their personal information
Precondition	The student is logged in
Main Flow	1. Student selects View Personal Information 2. System identifies the current student 3. System retrieves the student's personal information 4. System displays the information
Alternative Flow	If the information cannot be retrieved, the system displays an error message
Postcondition	The student's personal information is displayed

4.3.5. View Grades

Attribute	Description
Use Case	View Grades
Actor	Student / Teacher / School Administration
Goal	Allow authorized users to view grade information
Precondition	The user is logged in and has permission to view grades
Main Flow	1. User selects View Grades 2. System verifies the user's role 3. System retrieves the appropriate grade information 4. System displays the grades
Alternative Flow	If no grade information is available, the system displays a notification
Postcondition	The grade information is displayed

Access rules:

- + Student: view their own grades.
- + Teacher: view grades of students in assigned classes/subjects.
- + School Administration: view grades for management purposes.

4.3.6. View Academic Results

Attribute	Description
Use Case	View Academic Results
Actor	Student / Teacher / School Administration
Goal	Allow authorized users to view academic results
Precondition	The user is logged in
Main Flow	1. User selects View Academic Results 2. User selects a semester if necessary 3. System retrieves the academic results 4. System displays the results, average score, rank, and notes
Alternative Flow	If no academic result is available, the system displays a notification

Attribute	Description
Postcondition	Academic results are displayed

4.3.7. View Assigned Classes

Attribute	Description
Use Case	View Assigned Classes
Actor	Teacher
Goal	Allow a teacher to view the classes assigned to them
Precondition	Teacher is logged in
Main Flow	1. Teacher selects View Assigned Classes 2. System retrieves the assigned classes 3. System displays the class list
Postcondition	The assigned class list is displayed

4.3.8. View Students by Class

Attribute	Description
Use Case	View Students by Class
Actor	Teacher
Goal	Allow a teacher to view students in an assigned class
Precondition	Teacher is logged in and has an assigned class
Main Flow	1. Teacher selects a class 2. System retrieves the students in that class 3. System displays the student list
Postcondition	The student list is displayed

4.3.9. View Assigned Subjects

Attribute	Description
Use Case	View Assigned Subjects
Actor	Teacher
Goal	Allow a teacher to view the subjects assigned to them
Precondition	Teacher is logged in
Main Flow	1. Teacher selects View Assigned Subjects 2. System retrieves the assigned subjects 3. System displays the subject list
Postcondition	The assigned subject list is displayed

4.3.10. Enter Grades

Attribute	Description
Use Case	Enter Grades
Actor	Teacher
Goal	Allow a teacher to enter grades for students
Precondition	Teacher is logged in and is assigned to the corresponding class and subject
Main Flow	1. Teacher selects a class 2. Teacher selects a subject 3. Teacher selects a student 4. Teacher enters the grade 5. System validates the grade 6. System saves the grade 7. System displays a success message
Alternative Flow	If the grade is invalid, the system displays an error message and does not save the grade
Postcondition	The grade is successfully stored

4.3.11. Update Grades

Attribute	Description
Use Case	Update Grades
Actor	Teacher
Goal	Allow a teacher to update an existing grade
Precondition	Teacher is logged in and has permission to manage the grade
Main Flow	1. Teacher selects the grade to update 2. Teacher enters the new grade 3. System validates the new grade 4. System updates the grade 5. System displays a success message
Alternative Flow	If the new grade is invalid, the system displays an error message and does not update the grade
Postcondition	The grade is successfully updated

4.3.12. Manage Students

Attribute	Description
Use Case	Manage Students
Actor	School Administration
Goal	Allow the administrator to manage student information
Precondition	School Administration is logged in
Main Flow	1. Administrator opens Student Management 2. Administrator selects an operation 3. System performs the selected operation 4. System displays the result
Includes	View Students, Add Student, Update Student, Delete Student
Postcondition	Student information is successfully managed

Add Student

- + Administrator selects Add Student.
- + Enters student information.
- + System validates the information.
- + System saves the student.
- + System displays a success message.

View Students

- + Administrator selects View Students.
- + System retrieves student information.
- + System displays the student list.

Update Student

- + Administrator selects a student.
- + Updates the student's information.
- + System validates the information.
- + System saves the changes.

Delete Student

- + Administrator selects a student.
- + Selects Delete Student.
- + System asks for confirmation.
- + Administrator confirms.
- + System deletes the student.

4.3.13. Manage Classes

Attribute	Description
Use Case	Manage Classes
Actor	School Administration
Goal	Allow the administrator to manage school classes
Precondition	School Administration is logged in
Main Flow	1. Administrator opens Class Management 2. Administrator selects an operation 3. System performs the selected operation 4. System displays the result
Includes	View Classes, Add Class, Update Class, Delete Class
Postcondition	Class information is successfully managed

4.3.14. Manage Subjects

Attribute	Description
Use Case	Manage Subjects
Actor	School Administration
Goal	Allow the administrator to manage subjects
Precondition	School Administration is logged in
Main Flow	1. Administrator opens Subject Management 2. Administrator selects an operation 3. System performs the selected operation 4. System displays the result
Includes	View Subjects, Add Subject, Update Subject, Delete Subject
Postcondition	Subject information is successfully managed

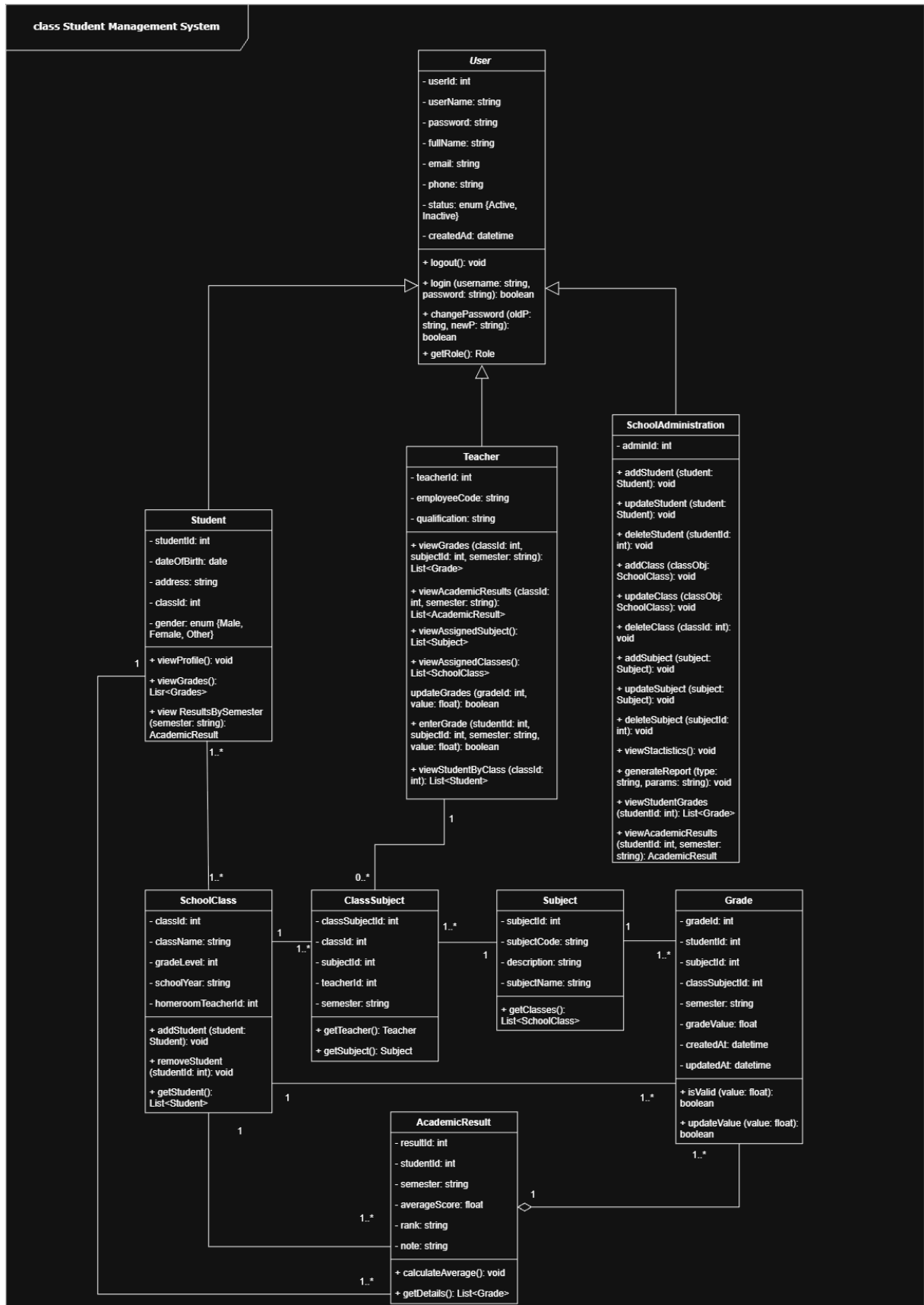
4.3.15. View Statistics

Attribute	Description
Use Case	View Statistics
Actor	School Administration
Goal	Allow the administrator to view statistics about students and academic results
Precondition	School Administration is logged in
Main Flow	1. Administrator selects Statistics & Reports 2. Selects the required statistics 3. Selects the class, subject, or semester if necessary 4. System retrieves and processes the data 5. System displays the statistics
Includes	Students by Class; Results by Class; Results by Subject
Postcondition	The requested statistics are displayed

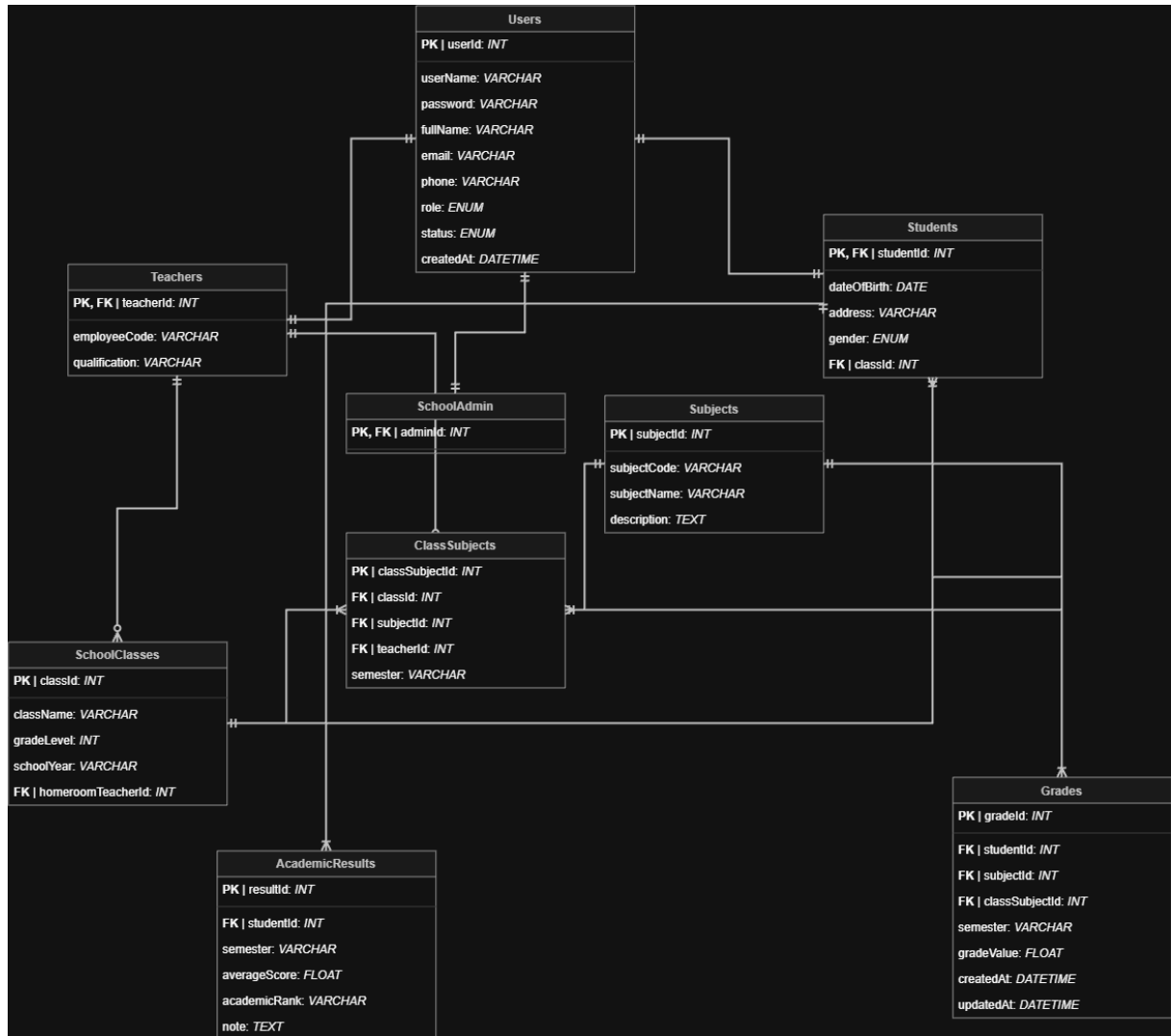
4.3.16. Generate Reports

Attribute	Description
Use Case	Generate Reports
Actor	School Administration
Goal	Allow the administrator to generate academic and management reports
Precondition	School Administration is logged in
Main Flow	1. Administrator selects Generate Reports 2. Selects the report type 3. System retrieves the required data 4. System processes and summarizes the data 5. System generates the report 6. System displays the generated report
Alternative Flow	If insufficient data is available, the system displays a notification
Postcondition	The report is successfully generated

5. Class Diagram



6. Data Model



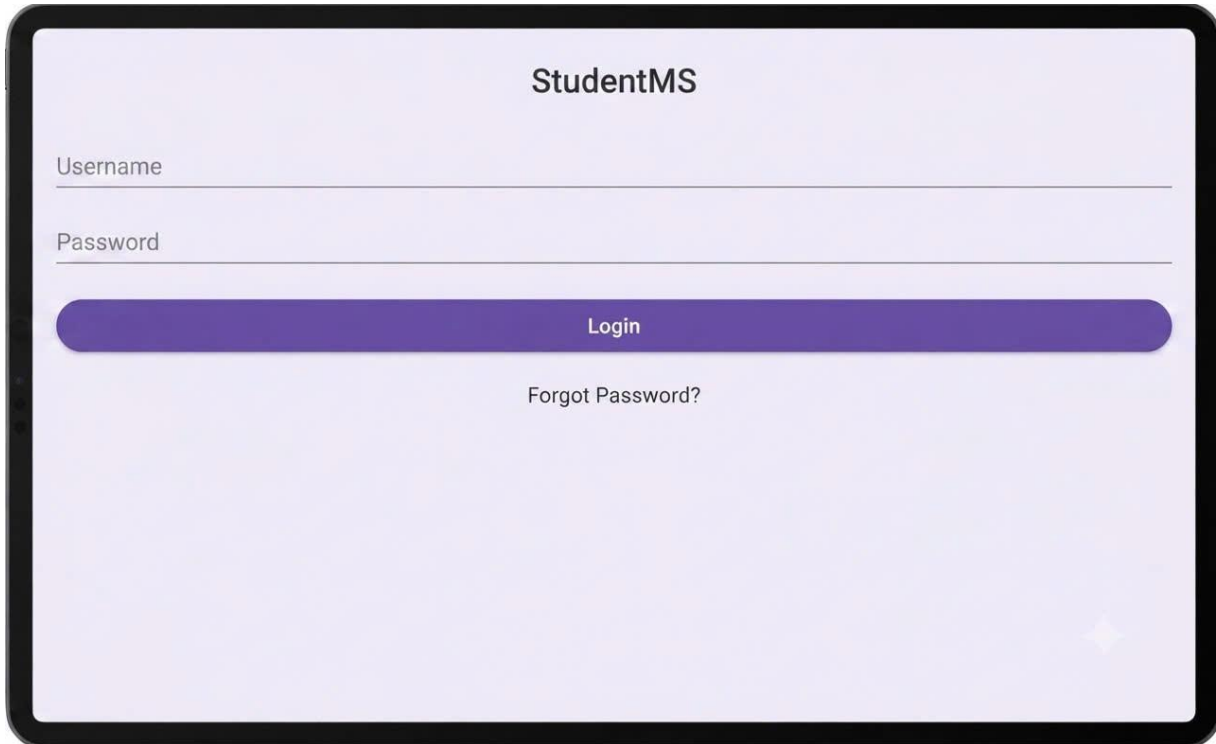
7. Interface Design Description

7.1. List of Screens

No.	Screen Name	Screen Type	Description
1	Login	Input Screen	Allows Student, Teacher, and School Administration to enter their username and password to access StudentMS.
2	Home	Main Screen	Displays the main functions of StudentMS according to the user's role and provides access to the system's management functions.
3	Student Management	Management Screen	Allows School Administration to view, add, update, and delete student information.
4	Class & Subject Management	Management Screen	Allows School Administration to manage classes and subjects, including class and subject information.
5	Grade Management	Input Screen	Allows teachers to enter, view, and update grades for students in their assigned classes and subjects.
6	Academic Results	Result Screen	Displays students' grades and overall academic results by semester.
7	Statistics & Reports	Report Screen	Displays statistics of students and academic results by class, subject, and semester.
8	Account Management	Management Screen	Allows users to change their password and log out of the StudentMS system.

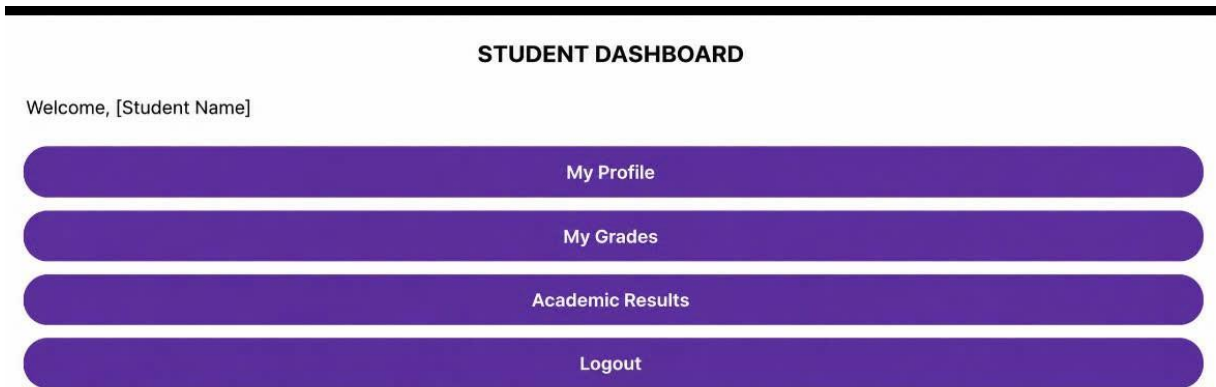
7.2. Screen Details

7.2.1. Login

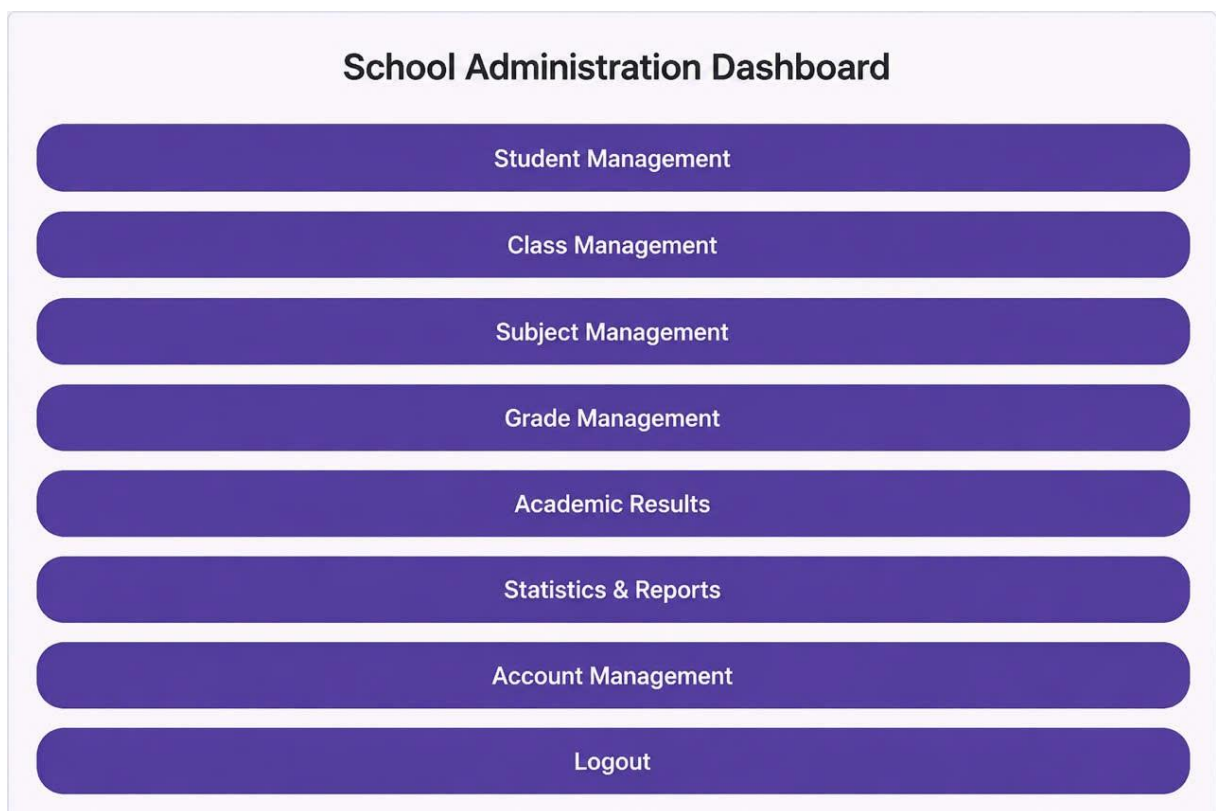
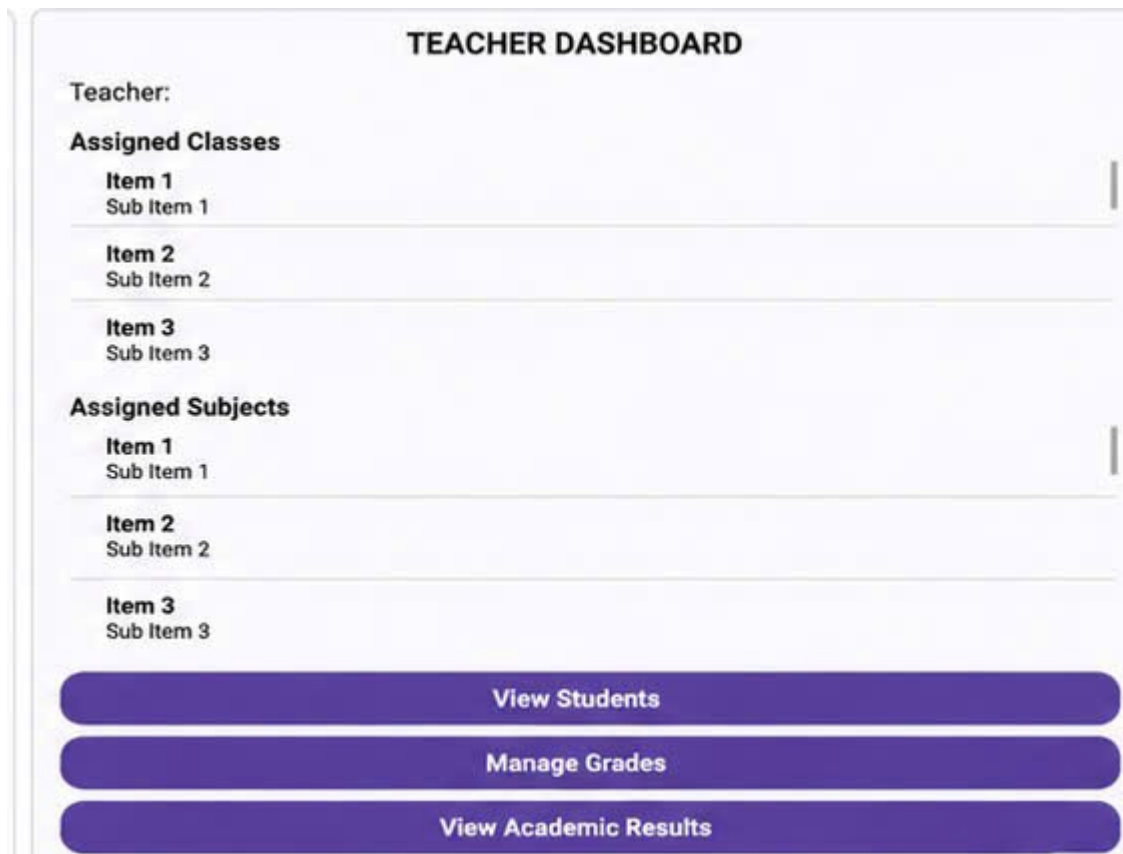


The login screen for StudentMS features a light purple background with a dark purple header bar containing the text "StudentMS". Below the header, there are two input fields: "Username" and "Password", each with a dark purple underline. A large, rounded dark purple button labeled "Login" is centered below the input fields. Below the button, the text "Forgot Password?" is displayed in a smaller font. The entire screen is framed by a thick black border.

7.2.2. Home



The student dashboard has a light purple background and a dark purple header bar with the text "STUDENT DASHBOARD". Below the header, the text "Welcome, [Student Name]" is displayed. Below this, there are four horizontal, rounded dark purple buttons stacked vertically, labeled "My Profile", "My Grades", "Academic Results", and "Logout".



7.2.3. Student Management

STUDENT MANAGEMENT

Student ID

Student ID

Full Name

Full Name

Date Of Birth

Date Of Birth

Item 1

Address

Item 1

Add

Update

Delete

Student List

Item 1

Sub Item 1

Item 2

Sub Item 2

7.2.4. Class & Subject Management

CLASS MANAGEMENT

Class ID

Class Name

Grade Level

School Year

Homeroom Teacher

Add

Update

Delete

SUBJECT MANAGEMENT

Subject ID

Subject Code

Subject Name

Description

Add

Update

Delete

7.2.5. Grade Management

GRADE MANAGEMENT

Item 1

Item 1

Item 1

Student ID

Student Name

Item 1

Grade

Item 1

Save Grade

Update Grade

Grade List

Item 1

Sub Item 1

Item 2

Sub Item 2

Item 3

Sub Item 3

7.2.6. Academic Results

ACADEMIC RESULTS

Student ID

Item 1

View Academic Result

Item 1

Sub Item 1

Item 2

Sub Item 2

Item 3

Sub Item 3

Item 4

Sub Item 5

Item 5

Sub Item 6

Item 7

Sub Item 7

Item 8

Sub Item 8

Item 9

Sub Item 9

Item 10

Sub Item 10

Average Score:

Rank:

Note:

STATISTICS & REPORTS

Students By Class

Results By Class

Results By Subject

Generate Report

STATISTICS

Total Students: 0

Excellent: 0

Good: 0

Average: 0

7.2.8. Account Management

ACCOUNT MANAGEMENT

Username:

Current Password:

New Password:

Confirm New Password:

Change Password

Logout